

Vasu Manocha

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SUMMARY

Software Engineer with experience building **data pipelines**, **scalable cloud backends**, and **LLM-powered platforms** in production. Architected serverless AWS infrastructure and data processing workflows at Apexon, supporting tools used by **2,000–3,000 HR professionals across 4 countries**. Strong in Python, SQL, and distributed data systems. Seeking Data/Software Engineering roles in Canada.

WORK EXPERIENCE

Software Engineer Jan 2025 – Mar 2026
Apexon

- Built and shipped the **JD Optimizer Agent** end-to-end: LLM-powered platform adopted by **2,000–3,000 HR professionals across 4 countries** to automate job description rewrites, skill extraction, bias detection, and tone calibration.
- Architected a **serverless AWS data backend** (Lambda, API Gateway, S3, DynamoDB) with FastAPI; designed ETL workflows and reusable REST API patterns supporting multiple AI tools simultaneously at scale.
- Delivered the **Interview Prep Agent** from design to deployment; led as **Squad Leader** for the Data Fellowship 2025 cohort – coordinated technical planning and an 8-week agile program for 10+ participants.

Undergraduate Teaching Assistant Jan 2023 – Dec 2024
Memorial University of Newfoundland

- Supported 100+ students per semester across CS and Mathematics courses; held weekly office hours and provided written feedback on assignments.

Undergraduate Research Assistant May 2023 – Sep 2023
Memorial University of Newfoundland

- Co-authored a **novel theorem** on the Cops & Robbers problem on directed circulant graphs, producing original proofs for academic publication.

Software Developer Intern Jun 2021 – Jul 2021
Konverse.ai, India

- Developed enterprise chatbots in Python and Dialogflow, automating customer-support flows and integrating third-party APIs to reduce manual handling time.

PROJECTS

FitAccess – Global Gym Discovery Platform 2024–2026
· Built and launched a full-stack gym discovery app with geolocation search, real-time availability, and Google Maps API integration – reached **2,000+ active users**.
· Stack: **Next.js, TypeScript, FastAPI, PostgreSQL, Docker, Maps API**.

Whatchu Cooking? – Social Recipe App 2025
· Built a cross-platform mobile app for social recipe sharing and potluck planning in a 4-person agile team; implemented voice-to-text input, Firebase Cloud Storage, Cloud Firestore sync, and Firebase Auth. Stack: **Flutter, Dart, Firebase, Riverpod**.

2D Game Engine & Platformer (C++) 2025
· Engineered a complete 2D game engine from scratch: entity-component-system (ECS) architecture, real-time physics simulation, enemy AI, and an interactive level editor.

Selected Projects

- **Tely 10 Training Plan**: Free personalized 7-week race training generator – adapts weekly mileage and pace targets to each runner's current fitness. Stack: **JavaScript, Vite**.
- **eVerify Email Validator**: Combines regex parsing and live SMTP checks to detect format errors and non-existent domains in real time.
- **Lucy Voice Assistant**: Fully offline NLP-powered assistant using speech recognition and natural language command parsing; used internally at Apexon.
- **Face Recognition Model**: Python + OpenCV system with real-time identification; built for automated attendance tracking.

TECHNICAL SKILLS

- **Languages**: Python, TypeScript, JavaScript, Dart, Java, C++, SQL, HTML, CSS
- **Backend & APIs**: FastAPI, REST APIs, system design, data structures and algorithms, PostgreSQL, Redis, Postman
- **Cloud & DevOps**: AWS (Lambda, API Gateway, S3, DynamoDB), Docker, GitHub Actions (CI/CD), Git
- **Data & AI**: Apache Spark, ETL pipelines, LangChain, RAG pipelines, LLM prompt engineering, multi-agent systems, model evaluation, PostgreSQL, Redis
- **Frontend & Mobile**: Next.js, React, Flutter, Firebase (Auth, Firestore, Cloud Storage), Riverpod, responsive web design
- **AI Dev Tools**: Claude Code, Codex, Cursor, v0, Lovable, Bolt.new

EDUCATION

B.Sc. Computer Science & Applied Mathematics Dec 2025
Memorial University of Newfoundland, St. John's NL GPA: 3.86 / 4.00

ACHIEVEMENTS

- Dean's List – 2023–24, 2024–25 & 2025–26, Memorial University of Newfoundland (top 10% of Faculty of Science)
- Vedic Maths National 1st Rank (Aug 2021)